3DWEBSOFT INC. FOUNDATION

FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

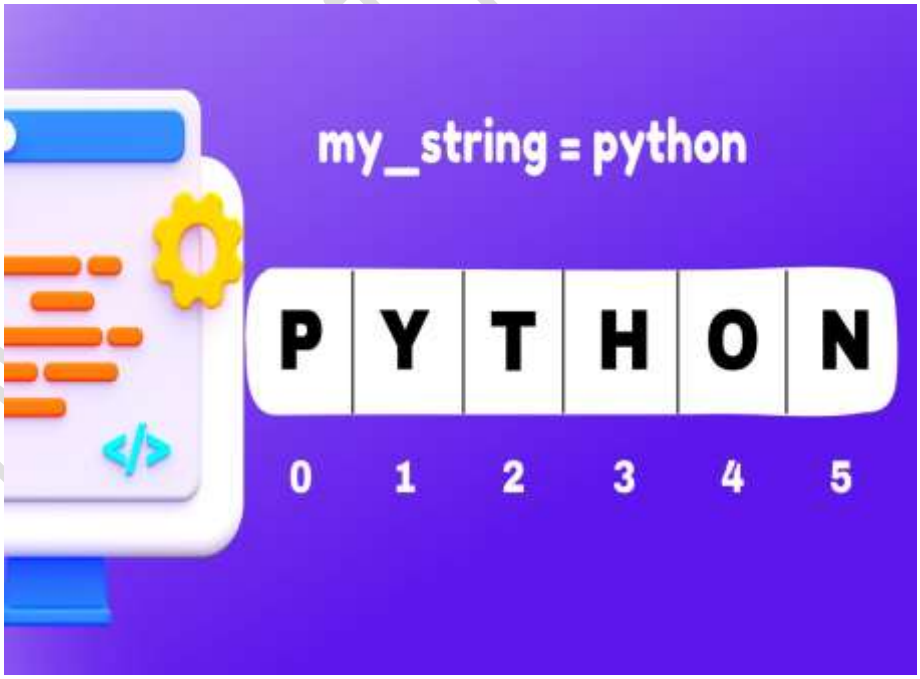
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Class 32: String Methods (Python)



DWEBSOFT FOUNDATION



Learning Objectives

By the end of this lesson, students will be able to:

- Understand what String Methods are.
- Use different string methods in Python.
- Convert text into uppercase and lowercase.

- Count letters and words.
- Replace words in a sentence.
- Split and join strings.
- Create simple text-based Python programs.

What is a String?

A **String** is a collection of characters enclosed in quotation marks.

Examples:

"Hello"

"Python"

"I love coding"

A string can contain:

- Letters
- Numbers
- Symbols
- Spaces

Example:

name = "Rahul"

city = "Hyderabad"

sentence = "Python is fun!"

What are String Methods?

String Methods are **special functions** that help us perform different operations on strings.

Think of them like **magic tools** that can:

- Change text
- Count letters
- Find words
- Replace words

DWEBSOFT FOUNDATION

- Remove spaces

Most Common String Methods

Method	Purpose
upper()	Converts to CAPITAL letters
lower()	Converts to small letters
title()	Makes First Letter Capital
capitalize()	Capitalizes first letter only
swapcase()	Changes upper to lower and lower to upper
replace()	Replaces words
count()	Counts letters/words
find()	Finds word position
startswith()	Checks beginning
endswith()	Checks ending
strip()	Removes spaces
split()	Splits sentence into words
join()	Joins words
len()	Counts total characters

1️⃣ upper()

Converts every letter into CAPITAL letters.

```
text = "python"


print(text.upper())
```

Output

PYTHON

DWEBSOFT FOUNDATION

2 lower()

Converts every letter into lowercase.

```
text = "PYTHON"


print(text.lower())
```

Output

```
python
```

3 title()

Capitalizes the first letter of every word.

```
name = "python programming"


print(name.title())
```

Output

```
Python Programming
```

4 capitalize()

Only the first letter becomes capital.

```
text = "hello world"


print(text.capitalize())
```

Output

```
Hello world
```

5 swapcase()

Changes uppercase into lowercase and lowercase into uppercase.

```
text = "Python"
```

DWEBSOFT FOUNDATION

print(text.swapcase())

Output

pYTHON

6 replace()

Replaces one word with another.

text = "I like mango"

print(text.replace("mango","apple"))

Output

I like apple

7 count()

Counts how many times a letter appears.

text = "banana"

print(text.count("a"))

Output

3

8 find()

Finds the position of a letter.

text = "Python"

print(text.find("t"))

Output

2

DWEBSOFT FOUNDATION

9️⃣ startswith()

Checks the starting word.

```
text = "Python"

print(text.startswith("Py"))
```

Output

True

🔟 endswith()

Checks the ending letters.

```
text = "Python"

print(text.endswith("on"))
```

Output

True

1️⃣1️⃣ strip()

Removes unwanted spaces.

```
text = "   Python   "

print(text.strip())
```

Output

Python

1️⃣2️⃣ split()

Breaks sentence into words.

```
text = "Python is easy"
```

DWEBSOFT FOUNDATION

print(text.split())

Output

['Python', 'is', 'easy']

1️⃣3️⃣ join()

Joins words together.

words = ["Python","is","fun"]

print(" ".join(words))

Output

Python is fun

1️⃣4️⃣ len()

Counts total characters.

text = "Python"

print(len(text))

Output

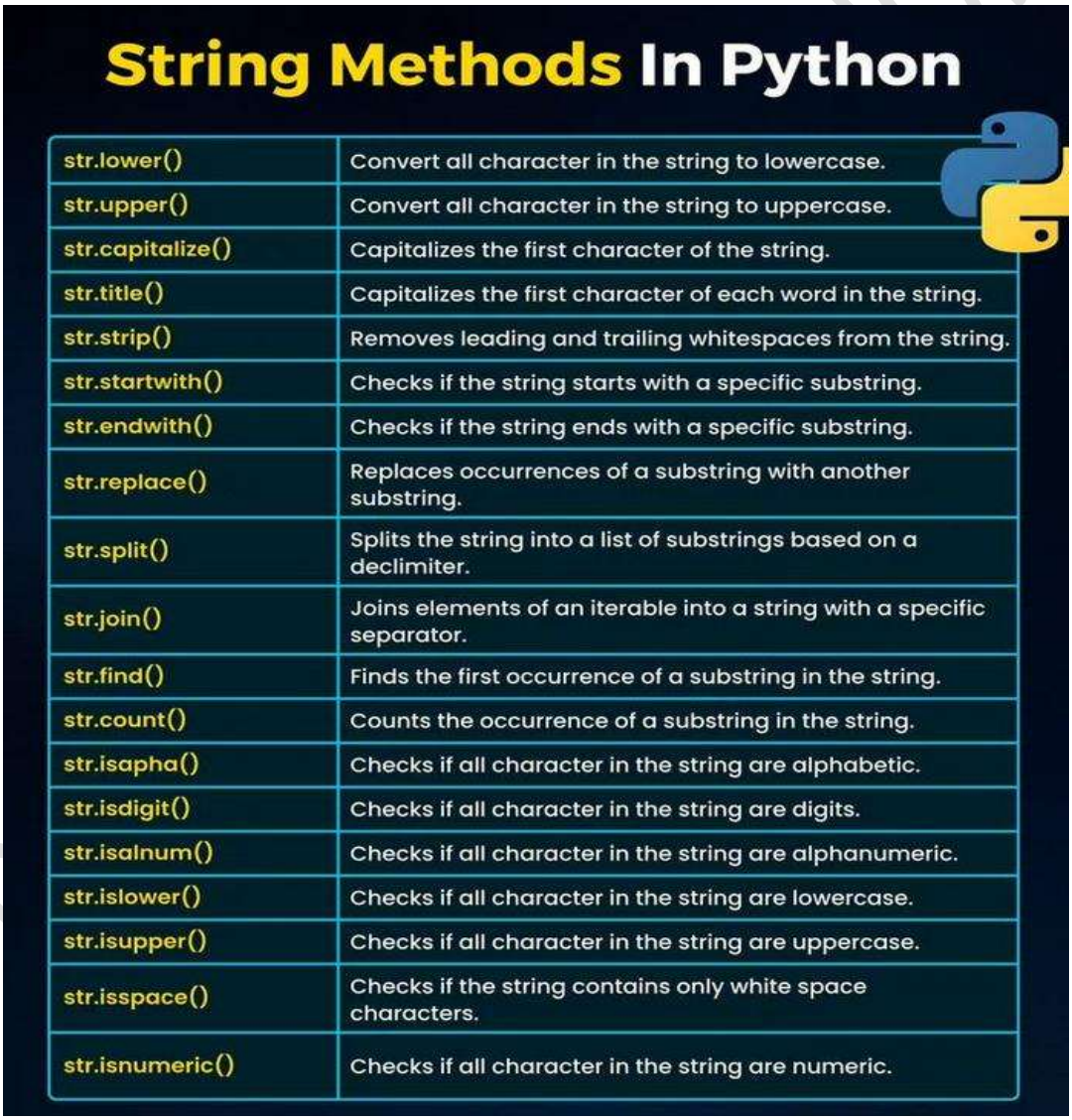
6

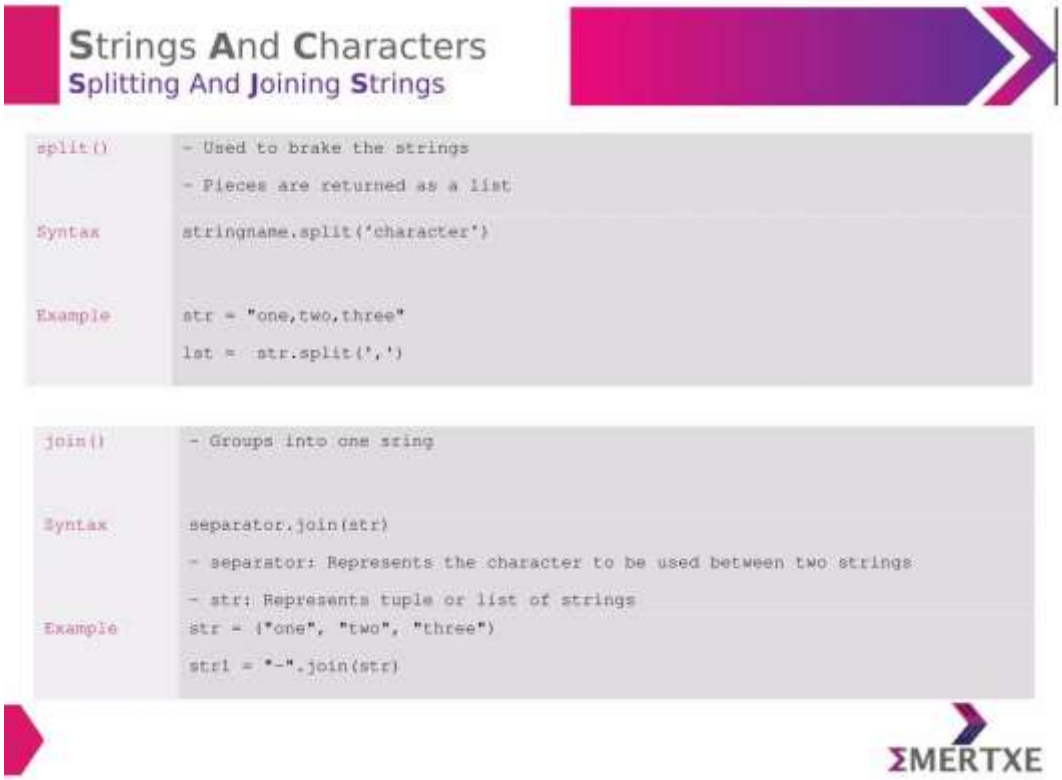
Memory Trick

Method	Easy Meaning
upper()	BIG letters
lower()	small letters
title()	Every Word Capital
replace()	Change word

DWEBSOFT FOUNDATION

Method	Easy Meaning
count()	Count letters
split()	Break sentence
join()	Join words
strip()	Remove spaces
find()	Find position





Practical Session

Practical Exercise 1

Convert to uppercase.

name = input("Enter your name : ")

print(name.upper())

Example

Input:

Rahul

Output:

RAHUL

Practical Exercise 2

DWEBSOFT FOUNDATION

Convert to lowercase.

```
city = input("Enter city : ")


print(city.lower())
```

Practical Exercise 3

Count letters.

```
word = input("Enter word : ")


print(word.count("a"))
```

Practical Exercise 4

Replace word.

```
sentence = input("Enter sentence : ")


print(sentence.replace("good","great"))
```

Practical Exercise 5

Remove spaces.

```
name = input("Enter name : ")


print(name.strip())
```

Practical Exercise 6

Split sentence.

```
sentence = input("Enter sentence : ")
```

DWEBSOFT FOUNDATION

```
print(sentence.split())
```

Practical Exercise 7

Find letter.

```
word = input("Enter word : ")


print(word.find("o"))
```

Practical Exercise 8

Check starting letters.

```
word = input("Enter word : ")


print(word.startswith("P"))
```

Practical Exercise 9

Check ending letters.

```
word = input("Enter word : ")


print(word.endswith("n"))
```

Practical Exercise 10

Join names.

```
names = ["Ali","Ravi","John"]


print(", ".join(names))
```

Output

Ali, Ravi, John

DWEBSOFT FOUNDATION

Mini Project: Student Name Formatter

```
name = input("Enter Student Name : ")


print("Upper :", name.upper())
print("Lower :", name.lower())
print("Title :", name.title())
print("Length :", len(name))
```

Example

Enter Student Name

rahul kumar

Upper : RAHUL KUMAR

Lower : rahul kumar

Title : Rahul Kumar

Length : 12

Classroom Activity

Ask students to type:

my school is wonderful

Use:

- upper()
- lower()
- title()

DWEBSOFT FOUNDATION

- split()
- replace()

Observe the outputs.

Practice Questions

1. What is a string?
2. What are string methods?
3. Which method converts text into uppercase?
4. Which method removes spaces?
5. Which method counts letters?
6. What is the use of replace()?
7. Which method splits a sentence?
8. What does join() do?
9. Which function finds the length of a string?
10. What is the difference between upper() and lower()?

Homework

Write a Python program to:

- Accept your full name.
- Print it in uppercase.
- Print it in lowercase.
- Print it in title case.
- Count the total characters.
- Replace your first name with **Student**.
- Split your name into separate words.

Summary

DWEBSOFT FOUNDATION

- A **String** is text enclosed in quotation marks.
- **String Methods** are built-in tools for working with text.
- upper() changes text to uppercase.
- lower() changes text to lowercase.
- title() capitalizes each word.
- replace() changes words.
- count() counts letters.
- find() finds positions.
- strip() removes spaces.
- split() divides text into words.
- join() combines words.
- len() counts the number of characters.

Fun Challenge

Write a Python program that asks the user to enter their favorite movie. Display:

- The movie name in **UPPERCASE**.
- The movie name in **lowercase**.
- The movie name in **Title Case**.
- The number of characters in the movie name.
- Whether the movie name starts with the letter **A** using startswith().

DWEBSOFT FOUNDATION